

Chapter 10: Determinants, the adjoint, and inverses Worksheet

1. Are each of the following statements true or false?

- (a) If A and B are $n \times n$ matrices with $\det(AB) = 0$, then $\det(A) = 0$ or $\det(B) = 0$.
- (b) If A is a 2×2 nonsingular matrix and $A = A^{-1}$, then $\det(A) = 1$.
- (c) If A, B, C are 4×4 matrices with $AB = AC$ and $\det(A) \neq 0$, then $B = C$.
- (d) If A is a 4×4 matrix with $\det(A) = 3$, then the rank of A is 3.

Solution.

- (a) True, just notice that $\det(AB) = \det(A)\det(B) = 0$.
- (b) False, Take $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. We have $A = A^{-1}$ but $\det(A) = -1$.
- (c) True, if $\det(A) \neq 0$ then A is nonsingular. Multiplying on the left by A^{-1} we get $B = A^{-1}AB = A^{-1}AC = C$.
- (d) False, if $\det(A) = 3 \neq 0$, so A is invertible. This means that A reduces to I_4 , so the rank of A is 4.

2. If A and B are 3×3 matrices with $\det(A) = 3$ and $\det(B) = -1$, compute

$$\det(A^3 B^2 A (A^T)^{-2} B)$$

Solution.

Using the multiplicative property of the determinant, we find that

$$\det(A^3 B^2 A (A^T)^{-2} B) = \det(A)^3 \det(B)^2 \det(A) \det(A^T)^{-2} \det(B) = \det(A)^4 \det(A^T)^{-2} \det(B)^3$$

Using the fact that $\det(A^T) = \det(A)$, we find that

$$\det(A)^4 \det(A^T)^{-2} \det(B)^3 = \det(A)^4 \det(A)^{-2} \det(B)^3 = \det(A)^2 \det(B)^3.$$

Now plugging in the values above, we get that

$$\det(A)^2 \det(B)^3 = 3^2(-1)^3 = -9.$$

3. For which value(s) of t does M have a nontrivial solution to $M\mathbf{x} = \mathbf{0}$?

$$M = \begin{bmatrix} t-3 & -2 \\ 2 & t+2 \end{bmatrix}$$

Solution.

$M\mathbf{x} = \mathbf{0}$ has a nontrivial solution exactly when it is not invertible, i.e. has zero determinant,

i.e. $(t-3)(t+2) - 2 \cdot (-2) = 0$. Now

$(t-3)(t+2) - 2 \cdot (-2) = t^2 - t - 2 = (t-2)(t+1)$ with roots $t = 2, -1$, so M has a nontrivial solution to $M\mathbf{x} = \mathbf{0}$ if and only if $t = 2, -1$.

4. Find all possible value(s) for $\det(A)$ if A is a 3×3 skew-symmetric matrix (ask your TA if you've forgotten this definition). What does this tell you about 3×3 skew-symmetric matrices?

Solution.

If A is skew-symmetric, then $A^T = -A$. Taking determinant of both sides, we have

$\det(A) = \det(A^T) = \det(-A) = (-1)^3 \det(A) = -\det(A)$. The only way for

$\det(A) = -\det(A)$ is if $\det(A) = 0$. Thus, all 3×3 skew-symmetric matrices are not

invertible. Fun side note: In general, if n is odd and A is skew-symmetric, then $\det(A) = 0$. If n

is even and A is skew-symmetric, then we can't say anything "easily" about the determinant

(Look up the Pfaffian if you're curious).

5. Suppose A and B are (unknown) 3×3 matrices, and A is row-equivalent to

$$M = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}, \text{ while } B \text{ is row-equivalent to } N = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Answer each of the following, if possible.

(a) What is the rank of A ? What is the rank of B ?

(b) What is the determinant of A ? What is the determinant of B ?

(c) How many solutions does $A\mathbf{x} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$ have? How many solutions does $B\mathbf{x} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$ have?

(d) Describe the solution set of $A\mathbf{x} = \mathbf{0}$. Describe the solution set of $B\mathbf{x} = \mathbf{0}$.

Solution.

(a) Since M is nearly in row-echelon form (just divide row 2 by 2 and row 3 by 4), we can see that there are 3 leading ones in a row-echelon form of A , so A has rank 3. Similarly, we can see that there are 2 leading ones in a row-echelon form of B , so B has rank 2.

(b) We can see that B is not invertible, since it has rank less than 3, so its determinant must be 0. However, because A is invertible (rank is 3), we can only determine that its determinant is nonzero. We cannot tell the actual determinant from M , since the row-reductions may have changed the value of the determinant.

(c) Since A is invertible, every linear system $A\mathbf{x} = \mathbf{b}$ has one (unique) solution. Since B is not invertible, we cannot tell whether or not $B\mathbf{x} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$ is consistent without the original B matrix, so this system has either no solution or infinitely many solutions.

(d) Since A is invertible, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$. We can continue reducing N to RREF: $RREF(B) = RREF(N) = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix}$, so if $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then $x_3 = t$, $x_2 = -3/2t$, and $x_1 = 3t$, so our solutions are $\mathbf{x} = \begin{bmatrix} 3t \\ -3/2t \\ t \end{bmatrix}$